01. Introduction to Cancer Treatment

Cancer treatment aims to remove, destroy, or control cancer cells while preserving healthy tissues as much as possible. The choice of treatment depends on several factors, including:

- **Type of cancer**
- **Stage of cancer**
- **Tumor size and location**
- **Patient's age**
- **Overall health**
- **Presence of other medical conditions**
- **Patient preferences**

Many patients receive a combination of treatments for the best outcome.

---

02. Goals of Cancer Treatment

The primary goals include:

- **Cure the cancer completely**
- **Control cancer growth**
- **Prevent cancer recurrence**
- **Relieve symptoms**
- **Improve quality of life**
- **Extend survival**

Treatment plans are individualized for every patient.

---

03. Surgery

Surgery is one of the oldest and most effective cancer treatments.

It involves removing the tumor along with some surrounding healthy tissue to reduce the risk of recurrence.

Common Uses

- **Breast cancer**
- **Colon cancer**
- **Lung cancer**
- **Kidney cancer**
- **Thyroid cancer**

- **Skin cancer**

Advantages

- **Removes localized tumors**
- **High cure rates in early-stage cancers**
- **Provides tissue for biopsy**
- **Immediate reduction of tumor burden**

Risks

- **Pain after surgery**
- **Bleeding**
- **Infection**
- **Scarring**
- **Temporary loss of function**
- **Longer recovery time**

---

04. Chemotherapy

Chemotherapy uses medicines to destroy rapidly dividing cancer cells.

These medicines circulate throughout the body and can treat cancers that have spread.

Methods of Administration

- **Intravenous (IV)**
- **Oral tablets**
- **Injection**
- **Infusion pump**

Common Side Effects

- **Hair loss**
- **Fatigue**
- **Nausea**
- **Vomiting**
- **Mouth ulcers**
- **Loss of appetite**
- **Increased risk of infection**
- **Low blood counts**

Most side effects improve after treatment ends.

---

05. Radiation Therapy

Radiation therapy uses high-energy rays to destroy cancer cells.

It damages the DNA of cancer cells, preventing them from growing.

Types

- **External Beam Radiation Therapy**
- **Internal Radiation Therapy (Brachytherapy)**

Benefits

- **Precise treatment**
- **Effective for localized cancers**
- **Can shrink tumors before surgery**
- **Can relieve pain in advanced cancer**

Possible Side Effects

- **Skin irritation**
- **Fatigue**
- **Swelling**
- **Difficulty swallowing (for head and neck cancers)**
- **Dry mouth**

---

06. Immunotherapy

Immunotherapy helps the body's immune system recognize and destroy cancer cells.

Unlike chemotherapy, it strengthens natural immune defenses.

Common Types

- **Immune checkpoint inhibitors**
- **CAR-T cell therapy**
- **Cancer vaccines**
- **Cytokine therapy**

Advantages

- **Long-lasting response**
- **Targets cancer cells specifically**

- **Fewer side effects in some patients**

Possible Side Effects

- **Fever**
- **Fatigue**
- **Skin rash**
- **Inflammation of organs**
- **Autoimmune reactions**

---

07. Targeted Therapy

Targeted therapy blocks specific molecules involved in cancer growth.

Unlike chemotherapy, it mainly attacks cancer cells while sparing normal cells.

Examples

- **EGFR inhibitors**
- **HER2 inhibitors**
- **BRAF inhibitors**
- **VEGF inhibitors**

Advantages

- **More precise treatment**
- **Fewer side effects**
- **Better outcomes for certain cancers**

---

08. Hormone Therapy

Some cancers grow because of hormones.

Hormone therapy blocks or lowers hormone levels.

Used For

- **Breast cancer**
- **Prostate cancer**

Methods

- **Hormone-blocking medicines**
- **Surgical removal of hormone-producing organs**

Side Effects

- **Hot flashes**
- **Fatigue**
- **Weight gain**
- **Mood changes**
- **Reduced bone density**

---

09. Stem Cell Transplant

Stem cell transplantation replaces damaged bone marrow with healthy stem cells.

It is mainly used for:

- **Leukemia**
- **Lymphoma**
- **Multiple myeloma**

Types

- **Autologous transplant**
- **Allogeneic transplant**

Risks

- **Infection**
- **Bleeding**
- **Graft-versus-host disease**
- **Organ complications**

---

10. Palliative Care

Palliative care focuses on improving comfort and quality of life.

It can be provided alongside active cancer treatment.

Services Include

- **Pain management**
- **Nutritional support**
- **Emotional counseling**
- **Psychological support**
- **Symptom control**

Palliative care is appropriate at any stage of cancer.

---

11. Combination Therapy

Many patients receive multiple treatments.

Examples include:

- **Surgery + Chemotherapy**
- **Surgery + Radiation Therapy**
- **Chemotherapy + Immunotherapy**
- **Chemotherapy + Targeted Therapy**
- **Radiation + Chemotherapy**

Combination therapy often provides better outcomes than a single treatment.

---

12. Personalized Treatment

Modern cancer treatment is increasingly personalized.

Doctors consider:

- **Genetic mutations**
- **Tumor biomarkers**
- **Cancer stage**
- **Overall health**
- **Previous treatments**

Personalized medicine improves treatment effectiveness while reducing unnecessary side effects.

---

13. Follow-Up Care

After treatment, patients require regular follow-up visits to:

- **Detect recurrence**
- **Monitor side effects**
- **Assess recovery**
- **Promote healthy lifestyle changes**

Follow-up schedules vary depending on cancer type and treatment received.

---

14. Lifestyle During Treatment

Patients are encouraged to:

- **Eat a balanced diet**
- **Stay hydrated**
- **Engage in light physical activity**
- **Get adequate sleep**
- **Avoid tobacco and alcohol**
- **Follow medication instructions**
- **Report new symptoms promptly**

Healthy habits support recovery and improve treatment outcomes.